

ASSET INVENTORY ASSIGNMENT

Question 1: what did I do?

For this assignment, I conducted a physical inventory of my home. I systematically went room by room, identifying items that could either **retrieve, store, or process data**. My inventory was organized into six categories: people, procedures, data, software, hardware, and network devices. After the inventory, I assessed each asset's risk level (**high, moderate, or low**) using the CIA triad criteria (**confidentiality, integrity, availability**).

People

For people, I included my immediate family: myself, my husband, and our two children. I also considered individuals who occasionally interact with our home, such as our babysitter, cleaner, and lawn service provider. Beyond physical presence, I acknowledged that online connections, such as Instagram followers, also represent an external "people" layer to my information environment. Since I do not use Facebook, that platform was excluded.

- **High risk:** Immediate family members (direct impact on privacy and safety).
- **Moderate risk:** Babysitter and cleaner (occasional trusted access).
- **Low risk:** Lawn service provider and Instagram followers (minimal access to house and data).

Procedures

I identified several household practices as procedural assets. These included parental controls on our kids' devices, shredding mail that contains PII, and our habit of setting the alarm system whenever we leave the house. Additional procedures were the use of a password manager on our phones, setting time limits on children's devices, and trash day schedules.

- **High risk:** Alarm system, password manager, rules for device use (if compromised, they expose us to both digital and physical risks).
- **Moderate risk:** Kid device procedures, shredding routines.
- **Low risk:** Trash Day schedule (publicly observable, minimal security relevance).

Data

Data assets included personal identification documents (birth certificates, passports, Social Security cards, driver's licenses), financial records (bank statements, receipts, tax returns), emails, health records (prescriptions, insurance documents), and household documents (mortgage agreements, warranties, utility contracts). I also considered resumes, photos, and videos, along with physical storage such as filing cabinets and safes.

- **High risk:** Identification documents, financial records, health records, emails, resumes, personal photos/videos.
- **Moderate risk:** Receipts, mortgage agreements, utility bills/contracts (contain some sensitive but not catastrophic information).
- **Low risk:** Warranties (minimal sensitive information).

Software

I included the software installed on home devices: communication apps (WhatsApp, Gmail), operating systems (Windows, iOS), productivity tools (Microsoft Office), alarm system app (HomePro), and social media (Instagram). Entertainment apps such as Netflix and Spotify were also listed.

- **High risk:** Gmail, WhatsApp, HomePro (access to communications and security).
- **Moderate risk:** Windows, iOS, Microsoft Office, Instagram (exposure risk but less critical).
- **Low risk:** Netflix and Spotify (entertainment only).

Hardware

For hardware, I documented personal and work laptops, smartphones, Apple Watch, tablets, USB drives, hard drives, and printers/scanners. I also considered baby monitors, smart TVs, filing cabinets, and safes.

- **High risk:** Laptops, smartphones, tablets, Apple Watch, storage devices (contain or access large volumes of sensitive data).
- **Moderate risk:** Printers/scanners, safes, filing cabinets (data-rich but locked/controlled). Smart TVs (less sensitive but still connected).
- **Low risk:** Not available in this group, as most hardware stores or transmits at least some data.

Network Devices

Finally, I assessed networking devices such as the Wi-Fi router, modem/Starlink, Wi-Fi extender, doorbell camera, baby monitor, Alexa, and Google Nest. These are particularly sensitive since they enable or extend connectivity and may record data.

- **High risk:** Wi-Fi router, modem/Starlink, doorbell camera, baby monitor, Alexa, Google Nest (all can be exploited for surveillance or access).
- **Moderate risk:** Wi-Fi extender (less critical but expands potential entry points).
- **Low risk:** Also, not available in this group, as most network devices stores or transmits at least some data.

Risk Classification

- **High risk assets (3):** Require strong confidentiality, integrity, and availability protections. Their compromise could impact privacy, finances, or safety. Examples include immediate family members, identity documents, financial/health records, laptops, smartphones, and core network devices.
- **Moderate risk assets (2):** Contain some sensitive data or involve occasional exposure but are less critical. Compromise could cause inconvenience but not total harm. Examples include babysitters/cleaners, receipts, mortgages, Windows/iOS, and smart TVs.
- **Low risk assets (1):** Hold minimal sensitivity. Their compromise would cause inconvenience, not major harm. Examples include the lawn service provider, trash day schedule, warranties, Netflix, and Spotify.

ID	ASSET NAME	DESCRIPTION	SYSTEM IDENTIFICATION
PEOPLE			
1	Akeem	My Husband/Significant Other	3
2	Tiwa	Me	3
3	Elliott	My 5-year-old Son	3
4	Ellie	My 2-year-old Daughter	3
5	Riley	Our Baby Sitter/ Service Provider	2
6	Margret	Our Cleaner/ Service Provider	2
7	Alvin	The guy who mows our lawn/ Service Provider	1
8	Instagram Followers	I have about 800 friends/followers on Instagram.	1
PROCEDURES			
9	Kid's device	Parental control on kid's devices	3
10	Mail Shredding	Shredding our mail that contains PII	2
11	Alarm system	Setting alarms when leaving the house	3
12	Rules	Setting time limits on kid's devices	2
13	Password Manager	Using a password manager on our phones	3
14	Trash day schedule	Trash day is weekly on Fridays.	1
DATA			
15	Birth Certificates	Personal Identification document	3
16	Passports	Personal Identification document	3
17	Social Security Card	Personal Identification document	3
18	Driver's License	Personal Identification document	3
19	Receipts	Financial Record	2
20	Monthly bank statements	Financial Record	3
21	Yearly Tax return	Financial Record	3
22	Emails	Contains a lot of sensitive and personal information	3

23	Prescriptions	Health information	3
24	Insurance document	Health information	3
25	Filing cabinets	Houses and keeps a lot of data. It is locked	2
26	Safes	Contains a lot of personal information and data like passports. Locked.	2
27	Resume	Digital file containing information about us	3
28	Photos	Physical information	3
29	Videos	Physical Information	3
30	Mortgage Agreement	Household document	2
31	Utility bills	Household document	2
32	Warranties	Household document	1
33	Utility Contracts	Household document	2
SOFTWARE			
34	WhatsApp	Communication software I use often to text and call	3
35	Gmail	Communication software for email	3
36	Windows	Operating system for my laptop	2
37	IOS	Operating system for my phone	2
38	Microsoft office	Software used for work, assignment and productivity in general	2
39	HomePro	Software used for alarm system	3
40	Netflix	Entertainment App	1
41	Spotify	Music App	1
42	Instagram	Social Media App where I post pictures	2
HARDWARE			
43	Personal Laptop	Used for school work	3
44	Office laptop and monitors	Used to work from home	3
45	Smartphone	Contains a lot of apps and personal information	3
46	Apple Watch	Contains apps and personal information including health data	3
47	Tablets	Contains a lot of apps and personal information	3
48	USB Drives	Contains a lot of old documents, resumes and pictures.	3
49	Hard Drives	Contains a lot of old documents, resumes and pictures.	3
50	Printer/Scanners	Used to print a lot of personal information which may be stored on it	3
51	Baby Monitor	Used to monitor our 2-year-old and can be tracked	3
52	Smart TVs	Has some software and applications on it alongside our log in information	2
53	Safes	Contains personal and confidential items. Locked	2
54	Filing cabinets	Houses a lot of data. Locked	2

NETWORKING DEVICES			
55	Wi-Fi Router	Transmits internet to our home	3
56	Modem/Star link	Used for internet on the go	3
57	Doorbell Camera	Linked to our alarm system but takes pictures of us whenever we open and close the door	3
58	Wi-Fi Extender	Extends the wi-fi network upstairs	2
59	Alexa	Alexa probably listens to all/most of our conversations and has information on us.	3
60	Google nest	Listens to most of our conversation and has information on us.	3
SYSTEM IDENTIFICATION LEGEND			
1		Low Risk Asset - Compromise may cause inconvenience but no major harm	
2		Moderate Risk Asset - Compromise is inconvenient but not catastrophic.	
3		High Risk Asset - Critical to protect. Requires strong CIA protection.	

Question 2: what were the results?

The results of my asset inventory showed that my home environment is full of information assets across all categories—people, procedures, data, software, hardware, and network devices. I identified immediate family members and occasional service providers as key people assets, since they interact with or manage data and devices. For procedures, I noted habits such as shredding mail, setting alarms, using parental controls, and managing passwords. My data assets included sensitive documents like passports, birth certificates, Social Security cards, tax returns, bank statements, and health records, as well as digital items like emails, resumes, and photos. I also catalogued software in daily use, from operating systems and productivity tools to communication and entertainment apps. Hardware assets included laptops, smartphones, tablets, printers, storage drives, and IoT devices like smart TVs and baby monitors. Finally, network devices such as the Wi-Fi router, modem, extenders, and smart assistants (Alexa, Google Nest) formed the backbone of connectivity.

When I grouped these assets by risk, I found that high-risk items included identity documents, laptops, smartphones, email accounts, the Wi-Fi router, and immediate family members themselves. These require strong confidentiality, integrity, and availability protections because compromise would directly impact privacy, finances, or physical safety. Moderate-risk assets included service providers, children's devices, receipts, mortgages, smart TVs, and filing cabinets, which are items that hold some sensitive data but are less critical. Low-risk assets, such as warranties, trash schedules, or entertainment apps, had minimal sensitivity and would only cause inconvenience if compromised.

How Inventory translates into Attack Surface

This inventory translates directly into my home's attack surface, which is the total set of entry points an attacker could exploit. People expand the surface through human error or social engineering, since a babysitter, child, or even an online connection could be manipulated to reveal sensitive information. Procedures form another layer: while practices like shredding mail or using password managers shrink the attack surface, lapses in those same routines create new vulnerabilities. Data is at the core of it all, since identity documents, financial records, and emails are the ultimate targets of many attacks.

Software and hardware broaden the surface by multiplying the ways an attacker can gain access - whether through outdated operating systems, malicious apps, or stolen devices. Network devices like routers, cameras, and smart assistants stretch the attack surface even further, since they connect the household to the internet and can be exploited remotely. In short, my inventory revealed that my home's attack surface is much wider than I initially realized, and managing it requires a combination of strong technical controls, secure procedures, and ongoing awareness from everyone in the household.

Question 3: What did I learn?

From this exercise, I learned that almost everything in a home can be an information asset and, by extension, part of the attack surface. Before doing the inventory, I thought of data mostly as documents and devices, but I realized people, procedures, and even everyday habits can create opportunities for attackers. For example, a babysitter who knows our Wi-Fi password or a child who clicks on a suspicious link are just as much a security consideration as the router itself.

I also learned how important it is to view risk through the lens of **confidentiality, integrity, and availability (CIA)**. A passport or Social Security card has high confidentiality requirements, while something like the alarm system must be available at all times. Thinking this way highlighted how different assets need different protections, and that some "low-risk" items can still act as steppingstones to compromise higher-value assets.

From a cybersecurity perspective, the biggest lesson was that my home's attack surface is much broader than I assumed. Smart devices, cloud apps, and even old hard drives all create potential entry points. To reduce this surface, I need to apply controls like strong authentication, regular updates, segmentation of IoT devices, and better physical protections for documents. Most importantly, I now see that cybersecurity is not just about technology but about **people, routines, and awareness**, because an attacker only needs one weak link to succeed.